ExpNo -14 Experiment to understand feature scaling

Program:

```
import numpy as np
import pandas as pd
df = pd.read_csv('pre-process_datasample.csv')
df
df.head()
# Fill missing Country values
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])
features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values
from sklearn.impute import SimpleImputer
age = SimpleImputer(strategy='mean', missing_values=np.nan)
Salary = SimpleImputer(strategy='mean', missing_values=np.nan)
age.fit(features[:, [1]])
Salary.fit(features[:, [2]])
features[:, [1]] = age.transform(features[:, [1]])
features[:, [2]] = Salary.transform(features[:, [2]])
features
from sklearn.preprocessing import OneHotEncoder
oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, [0]])
Country
final_set = np.concatenate((Country, features[:, [1, 2]]), axis=1)
final_set
from sklearn.preprocessing import StandardScaler
```

```
sc = StandardScaler()
sc.fit(final_set)
feat_standard_scaler = sc.transform(final_set)
feat_standard_scaler
from sklearn.preprocessing import MinMaxScaler
mms = MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set)
feat_minmax_scaler = mms.transform(final_set)
feat_minmax_scaler
```

Output:

```
array([[1.        , 0.        , 0.        , 0.73913043, 0.77419355],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 0.        , 1.        , 0.13043478, 0.19354839],
       [0.        , 0.        , 1.        , 0.47826087, 0.41935484],
       [0.        , 1.        , 0.        , 0.50543478, 0.32258065],
       [1.        , 0.        , 0.        , 0.34782609, 0.43145161],
       [0.        , 0.        , 1.        , 0.50543478, 0.12903226],
       [0.        , 0.        , 1.        , 0.91304348, 1.        ],
       [0.        , 1.        , 0.        , 1.        , 0.43145161],
       [0.        , 0.        , 1.        , 0.43478261, 0.61290323]])
```